

PHYS 201: Lab 4

Objectives: The primary objective of the experiment is to determine the equations that give the voltage across a capacitor as a function of time as it discharges through different resistors. From them, the general equation showing how the discharge depends on the capacitance and the resistance will be derived.

Distance (m)	Longitude Vm+	Longitude Vm-	Transverse Vm+	Transverse Vm-
0.16	0.312	-0.291	1.52	-1.524
0.20	0.18	-0.161	0.62	-0.594
0.25	0.105	-0.087	0.292	-0.251
0.30	0.067	-0.049	0.172	-0.125
0.35	0.047	-0.036	0.11	-0.068
0.40	0.037	-0.014	0.079	-0.036
0.45	0.03	-0.006	0.059	-0.021
0.50	0.026	0	0.046	-0.012

Table 1: Lab 4 Data

Uncertainty in Datapoints

Distance was measured using a meter stick, and the uncertainty is half of the smallest division:
 $\delta d = 0.0005\text{m}$

The uncertainty for all Vm measurements is: $\delta = 0.001$, and since we average Vm+ and Vm- ($f = \frac{V_{m+} - V_{m-}}{2}$) to get Vm the uncertainty using differential error analysis is as follows:

$$\delta V_m = \sqrt{\left(\frac{\partial f}{\partial x} \delta\right)^2 + \left(\frac{\partial f}{\partial y} \delta\right)^2} = \sqrt{\left(\frac{1}{2} * 0.001\right)^2 + \left(-\frac{1}{2} * 0.001\right)^2} = 0.001 * \sqrt{\frac{1}{2}}$$

$$= \pm 0.000707$$

By averaging the data we obtain Table 2 which we will graph to find the slope of the trendline

Distance (m)	Longitude Vm	Transverse Vm
0.16	0.3015	1.522
0.2	0.1705	0.607
0.25	0.096	0.2715
0.3	0.058	0.1485
0.35	0.0415	0.089
0.4	0.0255	0.0575
0.45	0.018	0.04
0.5	0.013	0.029

Table 2: Calculated Data

Using Python, Matplotlib, and Numpy I graphed Vm vs Distance in log10-log10 for both cases. To weigh the trendline towards the points where the distance was largest I used the w parameter in np.polyfit. Below is the code I used to fit a line to both graphs. there are 8 weights since there are 8 data points.

```
(b , a) = np.polyfit(np.log(x), np.log(y), 1, w=[1, 1, 1, 1, 1, 2, 2, 2])
```

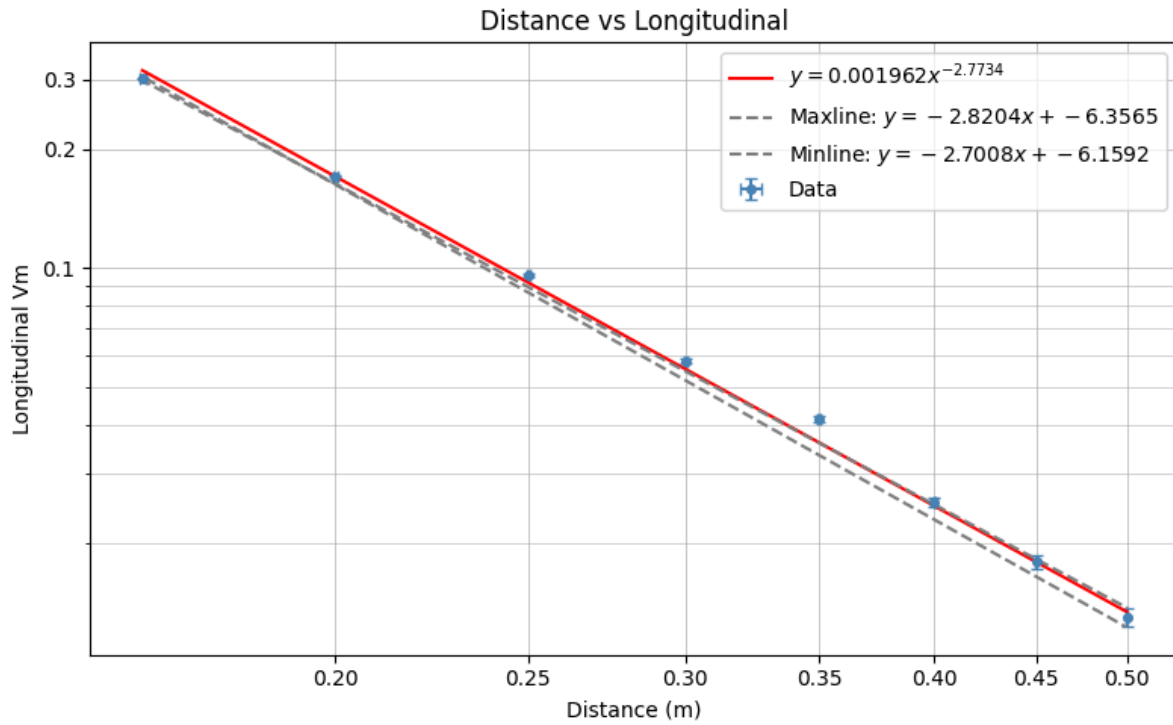


Figure 1: Graph 1

Numpy fit a power function with the equation: $y = 0.001962x^{-2.7734}$. Therefore the slope is: $m = -2.7734$

The equation for the field due to the north pole is: $B_n = \frac{m}{x^2(1-l/2x)^2}$. Where x is the distance and l is the size of the magnet. I'll average the difference of V_{m+} and V_{m-} for all points to get $l = 0.020\text{m}$. Plugging in m and l gives a function in terms of x :

$$B_n(x) = -\frac{2.7734}{x^2(1-0.020/2x)^2} = -\frac{2.7734}{(x-0.01)^2}$$

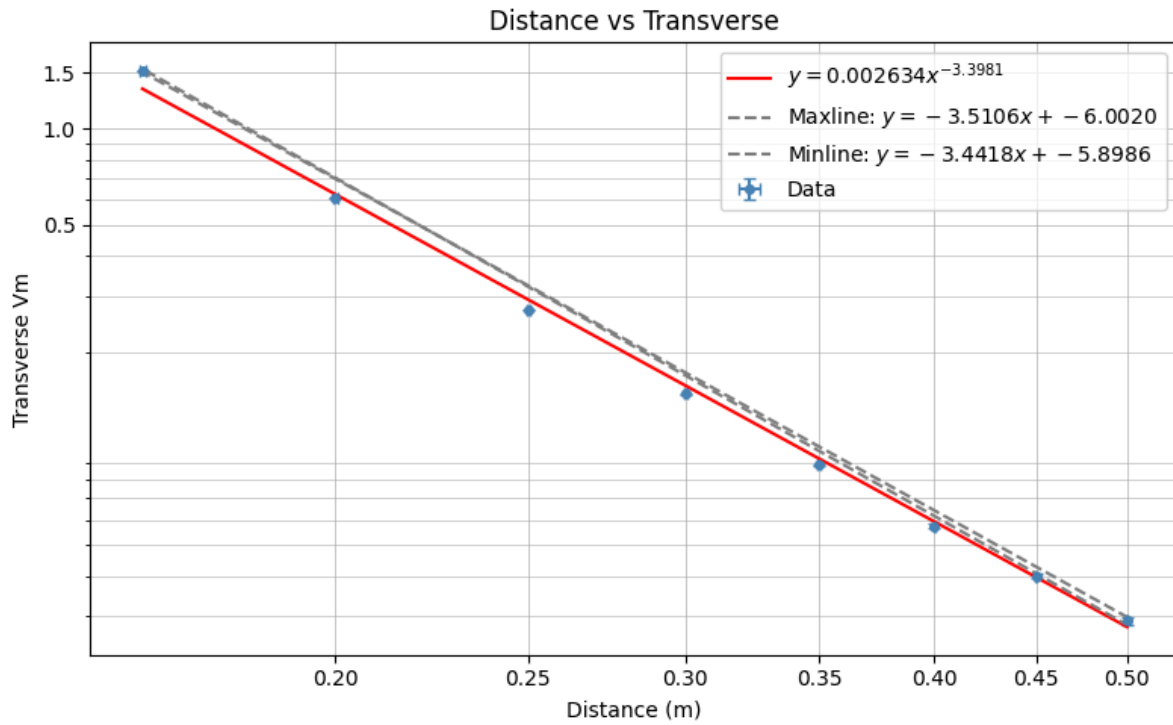


Figure 2: Graph 2

Numpy fit a power function with the equation: $y = 0.002634x^{-3.3981}$. Therefore the slope is: $m = -3.3981$

The equation for the field due to the south pole is: $B_s = -\frac{m}{x^2(1+l/2x)^2}$. $B_s(x) = -\frac{3.3981}{x^2(1+0.020/2x)^2} = -\frac{3.3981}{(x-0.01)^2}$

Adding B_n and B_s together gives the equation for the magnetic field where x varies:

$$B = B_n + B_s = -\frac{6.1715x^2 - 0.012494x + 0.00061715}{(x^2 - 0.0001)^2}$$